



New technology to block facial authentication fraud

Fujitsu Laboratories Ltd. has announced the development of a facial recognition technology that uses conventional cameras to successfully identify efforts to spoof authentication systems. This includes impersonation attempts in which a person presents a printed photograph or an image from the Internet to a camera.

Conventional technologies rely on expensive devices like near-infrared cameras to identify telltale signs of forgery, or the user is required to move their face from side to side, which remains difficult to duplicate with a forgery. This leads to increased costs, and the need for additional user interaction

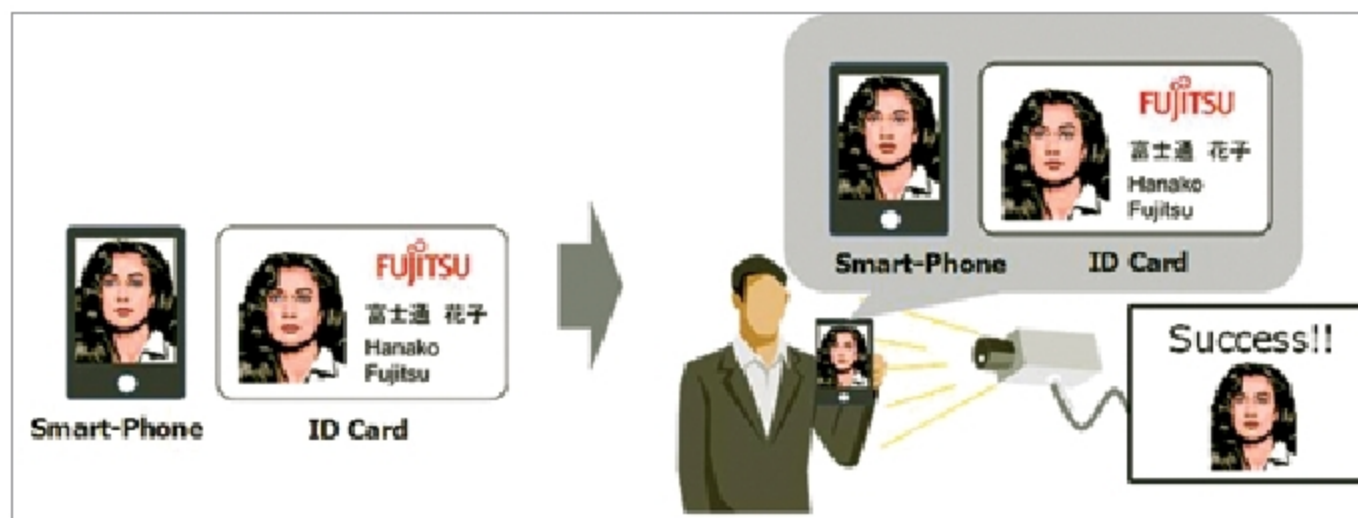
slows the authentication process. To tackle these challenges, Fujitsu has developed a forgery feature extraction technology that detects the subtle differences between an authentic image and a forgery.

This new technology ultimately makes it possible to prevent impersonation with forgeries using only face images taken at the time of authentication, thereby enhancing security without sacrificing the convenience of face authentication and contributing to the DX (digital transformation) of operations with improved personal authentication technologies.

The face image captured by the camera is separated into

various elements that exhibit the characteristic features of forgery, such as reflection elements and shape elements.

Next, image processing technology is used to digitise the characteristic features of forgery for each of the separated elements, and the characteristics of each element are combined to generate a characteristic for judgment. This makes it possible to identify counterfeits without information based on user operations.



Attempt to spoof ID card authentication with a photo to impersonate other (Credit: Fujitsu)

Google turns Android phones into earthquake detectors

Google has added an earthquake alert system in California, saying that it has been working on letting Android-powered smartphones double as tremor detectors. Android phones will receive warnings triggered by a 'ShakeAlert' earthquake early-warning system implemented on the West Coast by the US Geological Survey and partners.

ShakeAlert uses signals from hundreds of seismometers across the state to trigger warning messages that "an earthquake has begun and shaking is imminent," according to the system's website. "We saw an opportunity to use Android to provide people with timely, helpful earthquake information when they search, as well as a few seconds warning to get

themselves and their loved ones to safety if needed," principal software engineer Marc Stogaitis said in a blog post.

Smartphones are typically equipped with tiny accelerometers that sense movement and can catch shaking caused by earthquakes, according to Stogaitis. "This means your Android phone can be a mini seismometer, joining millions of other Android phones out there to form the world's largest earthquake detection network," Stogaitis said.

Phones that detect what might be earthquake activity can automatically send a signal to a data centre, where computers quickly determine whether a quake is happening, according to Google.

YouTube to use ML to catch videos that violate age restrictions

Google-owned YouTube is working towards expanding its machine learning (ML) technology to automatically catch videos that may require age restrictions, reports The Verge. This means that it would become difficult for underage viewers

to watch age-restricted videos.

People watching YouTube videos on third-party sites will also be redirected to YouTube to sign in and verify age. "We're expanding YouTube's machine learning technology